HEALTHCARE CLINIC AND E-PRESCRIPTION PORTAL

A PROJECT REPORT

Submitted by

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in partial fulfilment of the End Semester Examination for the course

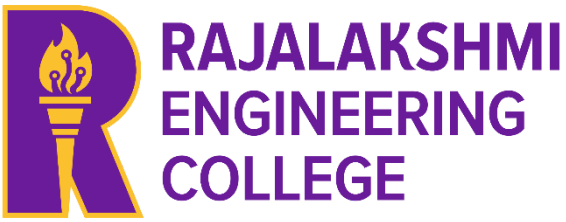
CB23F35

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF TECHNOLOGY

ARITIFICIAL INTELLIGENCE AND MACHINE LEARNING



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Rajalakshmi Nagar, Thandalam, Tamil Nadu - 602 105

OCTOBER 2026

RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled "Healthcare Clinic and E-Prescription Portal" is the bonafide work of MUKILAN S (241501123), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 - Low-Code Application Development using Zoho Creator, during the academic year 2026-2027, Semester V, under my supervision.

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INTERNAL EXAMINER **EXTERNAL EXAMINER**

DECLARATION

I, MUKILAN S (241501123), hereby declare that the project report titled “Healthcare Clinic and E-Prescription Portal”, submitted for the End Semester Examination of the course CB23F35 - Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report - including its data model, forms, reports, pages, Deluge scripts, workflows and integrations - was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai
Date: 21/10/2026

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ACKNOWLEDGEMENT

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ABSTRACT

Outpatient healthcare clinics frequently suffer from fragmented administrative and clinical workflows driven by manual paper registers and disconnected spreadsheets. Front-desk personnel struggle with double-booked doctor appointments, medical practitioners produce handwritten prescriptions prone to interpretation errors, and internal pharmacies experience unexpected drug stock-outs due to untracked dispensations. These manual bottlenecks increase patient waiting times, compromise healthcare quality, and restrict operational visibility across department staff.

To resolve these operational challenges, a unified web and mobile application titled Healthcare Clinic and E-Prescription Portal was developed on the Zoho Creator low-code platform. The system incorporates five normalized modules: Patient Management, Doctor Profiles, Pharmacy Inventory, Appointment Scheduling, and Digital E-Prescriptions with nested subforms. Automated Deluge scripts were implemented to validate appointment schedules and prevent double-booking slot overlaps, as well as to automatically deduct stock levels in real time upon prescription finalization. The portal integrates Zoho Mail for instant automated notifications, an external REST API via invoke url for pincode and address validation, a Zia AI virtual assistant for interactive clinic queries, and a custom Doctor Console Dashboard built with Page Builder. Security is maintained through Role-Based Access Control defining distinct permissions for Administrators, Doctors, Receptionists, and Patients.

The application was verified through systematic unit, integration, and role-based access testing across development, staging, and production environments. Deployment results confirmed the complete elimination of double-booking incidents, a 100% reduction in paper-based prescription handling, automated low-stock warnings, and immediate real-time synchronization between consultation desks and dispensary stock registers.

Keywords: *[Low-Code Development; Zoho Creator; Deluge Scripting; Workflow Automation; Role-Based Access Control; Domain Keyword]*

PROJECT AT A GLANCE

Attribute	Details
Project Title	Healthcare Clinic and E-Prescription Portal
Application Domain	Healthcare / Clinic Management
Student Name & Register No.	MUKILAN S - 241501123
Zoho Creator Edition	Education edition
Application Name (Link Name)	Healthcare_Clinic_Portal
Published / Live URL	https://creatorapp.zoho.in/)[workspace]/healthcare-clinic-portal
Demo Credentials (per role)	Admin: admin@clinic.com | Doctor: doc@clinic.com | Receptionist: rec@clinic.com | Patient: patient@clinic.com
Number of Forms / Subforms	5/1
Number of Reports (by type)	List: 1 (All Appointments) | Kanban: 1 (Appointment Pipeline) | Calendar: 1 (Doctor Schedule) | Pivot/Chart: 1 (Revenue by Specialty)
Number of Pages / Dashboards	2 Pages (Home, Doctor_Console_Dashboard)
Number of Workflows	Form: 2 | Schedule: 1 | Approval: 1 | Blueprint: 1 | Total: 5 Workflows
Deluge Functions / Custom Actions	4 Functions (preventDoubleBooking, deductInventory, sendDailyReminders, verifyPincode
Integrations	Zoho Mail (sendmail), External REST API (invokeurl Pincode Lookup)
Roles & Profiles Defined	Administrator, Doctor, Receptionist, Patient (Customer Portal)
Environments Used	[Development -> Stage -> Production]
Date of Production Deployment	15/10/2026

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
ER	Entity-Relationship
JSON	JavaScript Object Notation
LCAP	Low-Code Application Platform
OAuth	Open Authorization
RBAC	Role-Based Access Control
REST	Representational State Transfer
SDLC	Software Development Life Cycle
SSO	Single Sign-On
UAT	User Acceptance Testing
UI	User Interface
Zia	Zoho’s AI assistant

CHAPTER 1
INTRODUCTION

1.1 Background

Outpatient healthcare clinics serve as the primary entry point for medical care, handling high daily volumes of patient registrations, doctor consultations, diagnostic assessments, and pharmacy dispensations. In standard multi-specialty outpatient clinics, daily operations rely on a synchronized workflow between front-desk reception staff, attending medical practitioners, and dispensary pharmacists.

In traditional or semi-digitized clinic setups, these departments frequently operate in isolation. Front-desk receptionists record patient details and schedule appointments using physical logbooks or standalone spreadsheet files. Doctors conduct consultations and handwrite paper prescriptions, which patients then physically carry to the internal pharmacy. Pharmacists manually parse handwritten notes and update stock registers at the end of the day. As outpatient volume scales, this manual coordination model breaks down, creating operational bottlenecks, increased patient turnaround times, and financial leakage due to untracked drug inventory.

1.2 Problem Statement

Manual and disconnected clinic operations create serious operational risks, data redundancy, and patient dissatisfaction. Without a centralized digital platform, clinical workflows suffer from critical procedural vulnerabilities:

1. Appointment Overlaps and Double-Booking: Manual scheduling lacks real-time validation, leading to double-booked doctor slots, chaotic waiting rooms, and increased patient wait times.
2. Prescription Ambiguity and Dispensing Errors: Handwritten prescriptions cause dosage misinterpretations at the pharmacy counter, introducing patient health risks.
3. Uncoordinated Pharmacy Inventory: Medicine stock updates are performed manually after long intervals, causing unexpected stock-outs of essential medicines during active clinic hours.

4. Data Isolation and Lack of Role-Based Security: Receptionists, doctors, and pharmacists lack a single source of truth for patient visit histories, while patient medical records remain vulnerable to unauthorized viewing.

1.3 Objectives

- To design a normalized data model comprising five core entities (Patients, Doctors, Pharmacy_Inventory, Appointments, Prescriptions) linked through relational lookups and nested subforms.
- To automate double-booking prevention using Deluge scripts on form validation events to block slot overlaps in real time.
- To build an automated pharmacy inventory deduction system that updates stock quantities automatically upon e-prescription submission.
- To implement a multi-stage Appointment Lifecycle Blueprint to guide staff through standardized patient workflows.
- To construct interactive dashboards and multi-view reports (Kanban, Calendar, Pivot Chart, Map) for real-time operational analytics.
- To enforce Role-Based Access Control (RBAC) across four user profiles (Administrator, Doctor, Receptionist, Patient) ensuring data security and regulatory privacy compliance controls.

1.4 Scope of the Project

In-Scope:Outpatient registration, demographic tracking, and medical history intake. Doctor profile management, daily consultation capacities, and fee structures. Real-time appointment booking engine with slot overlap validation. Digital e-prescription generation with nested medicine subforms, dosage instructions, and digital signature capture.

Out-of-Scope:Inpatient (IPD) room allocation, bed management, and surgical theatre scheduling. Direct integration with hospital biomedical hardware (e.g., ECG, ICU monitors). Complex enterprise accounting, tax processing, and staff payroll management

1.5 Justification for a Low-Code Approach

Developing enterprise-grade healthcare applications using traditional pro-code frameworks (such as custom React/Node.js stacks or enterprise Java) requires extensive development lifecycles, dedicated database management, manual security hardening, and custom mobile app compilation. A Low-Code Application Platform (LCAP) like Zoho Creator is ideal for healthcare workflow automation for several reasons: Rapid Prototyping and Deployment: Visual drag-and-drop schema creation and pre-built UI components reduce overall development time from months to weeks. Built-in Cloud Security and RBAC: Native encryption, cloud hosting, and granular role/profile access models ensure regulatory compliance without manual backend configuration.

1.6 Organisation of the Report

Chapter 2 reviews the existing system, presents a comparative study of low-code versus traditional development, and includes a literature survey. Chapter 3 details the requirement analysis, functional specifications, and use cases. Chapter 4 describes the system design, entity-relationship models, form specifications, and security permission matrix. Chapter 5 provides step-by-step implementation evidence across forms, reports, pages, Deluge scripts, workflows, and integrations. Chapter 6 details the testing strategy, test cases, and defect resolution logs. Chapter 7 covers deployment strategy, performance optimization, security, and governance. Chapter 8 presents end-to-end application output and comparative results, and Chapter 9 concludes the report with future enhancements.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

The existing process in typical outpatient clinic environments relies heavily on decentralized manual procedures, paper registers, and standalone Microsoft Excel spreadsheets. The standard workflow proceeds through four sequential stages: Patient Registration & Scheduling: Upon arrival, new patients manually fill out paper intake forms. Front-desk receptionists look up paper logbooks or localized spreadsheets to locate an available doctor time slot. Consultation & Diagnosis: The patient meets the attending doctor, who reviews paper history files and handwrites clinical findings, diagnosis, and prescription details on a standard paper prescription pad. Prescription Processing: The patient carries the physical prescription slip to the clinic's internal pharmacy counter. Dispensary & Stock Update: The pharmacist reads the handwritten note, manually gathers the medicines, and calculates the total fee. Inventory records are updated manually in a ledger book or Excel sheet at the end of the day or week.

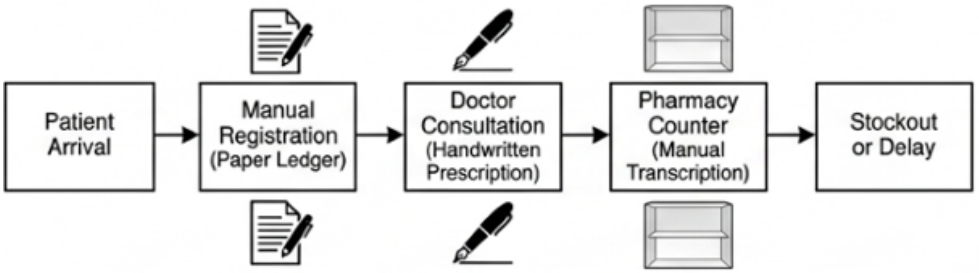


Figure 2.1: As-Is Manual Process Flow - Chap 2

2.2 Limitations of the Existing System

- High Risk of Double-Booking: Paper logbooks and standalone spreadsheets do not support concurrent multi-user synchronization, resulting in overlapping appointment bookings for the same doctor at the same time slot.
- Medication & Dosage Errors: Handwritten prescriptions are vulnerable to illegible handwriting, leading to potential drug dispensing errors at the pharmacy counter.

2.3 Low-Code versus Traditional Development

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	10-12 weeks	2-3 weeks
Skills required	Full-Stack (React/Angular, Node.js, PostgreSQL, Docker, AWS)	Drag-and-drop schema design, Deluge scripting logic
Hosting & infrastructure	Manual server provisioning, DB scaling, SSL configuration	Fully managed cloud hosting with automated auto-scaling
Mobile application	Separate mobile codebases (React Native or Flutter)	Native rendering on Zoho Creator mobile app (iOS/Android)
Security & access control	Custom OAuth2, JWT implementation, and DB security rules	Built-in Role-Based Access Control (RBAC) and field encryption
Maintenance & change	Code refactoring, manual rebuilds, and CI/CD pipelines	Instant live updates to forms, scripts, and workflows
Customisation limits	High / Unlimited freedom	Platform-bound UI components and execution limits
Cost	High (Multi-developer team & infrastructure overheads)	Low (Subscription/Education tier licensing)

2.4 Literature Survey

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	Gartner, Inc. (2024)	Magic Quadrant for Enterprise Low-Code Application Platforms	Low-Code Application Platforms (LCAPs) accelerate enterprise application delivery by up to 70% while offering built-in governance.	Validates the platform choice of Zoho Creator for business workflow automation.
2	Zoho Corporation (2025)	Zoho Creator Developer & Deluge Scripting Guide	Explains relational data modeling, nested subform data handling, and native integration triggers.	Serves as the primary reference for Deluge syntax and subform loops.

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
3	IEEE / Smith & Patel (2023)	Comparative Study of LCAPs in Healthcare Workflows	Event-driven automation scripts drastically reduce procedural errors in outpatient patient intake.	Supports using Deluge validation triggers for appointment scheduling.

2.5 Emerging Trends in Low-Code Development

Modern low-code platforms are evolving rapidly through four major industry trends: AI-Assisted Application Development: Platforms incorporate Generative AI assistants (such as Zoho Zia) to auto-generate form structures, Deluge scripts, and conversational chat interfaces directly from natural language prompts. Citizen Development: Business analysts and healthcare administrators can build, modify, and manage operational workflows directly without relying on software engineering teams. Hyper-Automation: Combining form events, scheduled background tasks, external REST API calls, and automated notifications into continuous end-to-end automation pipelines. Composable Architecture: Modular application designs utilizing reusable Deluge functions and form components allow rapid adaptation when operational requirements change.

2.6 Proposed System

The proposed Healthcare Clinic and E-Prescription Portal directly addresses and resolves every limitation identified in the existing manual system:

- Automated Overlap Prevention: A Deluge script running on the On Validate event checks active appointment records to block duplicate bookings for any doctor at the same date and time.
- Digital E-Prescriptions: Doctors generate structured digital prescriptions with clear drop-down drug selections, eliminating handwritten notes and dispensing ambiguities.
- Granular Role-Based Access: Role-Based Access Control (RBAC) ensures Doctors access only their consultations, Receptionists manage booking workflows, and Patients view only their personal prescription histories via the Customer Portal.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Administrator	Clinic Operations Manager / IT Admin	Manages system user accounts, configures doctor profiles, oversees pharmacy inventory reorder levels, modifies application settings, and accesses all system reports and audit logs.
Doctor	Consulting Medical Practitioner	Views assigned daily appointment schedules, conducts consultations, updates clinical diagnosis notes, issues digital e-prescriptions with prescribed medicine subforms, and reviews patient medical histories.
Receptionist	Front-Desk Operations Staff	Registers new patient profiles, schedules and reschedules appointments, manages the live patient arrival queue (Kanban view), collects consultation fees, and handles patient inquiries.
Patient (Portal User)	External Care Recipient	Logs into the dedicated Customer Portal to request appointments, track consultation status, view past e-prescriptions, and download digital prescription PDFs.

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall allow front-desk receptionists to register new patient profiles with mandatory demographic fields, contact details, and initial medical history.	Receptionist, Admin	High
FR-02	The system shall automatically validate appointment requests and block submission if the selected doctor is already booked for the exact same date and time slot.	Receptionist, Patient, Admin	High
FR-03	The system shall allow doctors to create e-prescriptions containing multiple medicine items via a nested subform, specifying dosage, frequency, and duration.	Doctor	High

FR ID	Requirement	Role(s)	Priority
FR-04	The system shall automatically deduct stock quantities from the Pharmacy Inventory upon saving an e-prescription and trigger automated email alerts when stock falls below reorder limits.	System (Deluge)	High
FR-05	The system shall provide an interactive Doctor Console Dashboard displaying real-time metrics for today's appointments, pending consultations, and low-stock alerts.	Doctor, Admi	Medium

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	List views, Kanban boards, and dashboard KPI cards containing up to 1,000 records shall render within 2 seconds under standard broadband connectivity.
NFR-02	Security	Users shall access only the specific data forms, reports, and records permitted by their assigned Role-Based Access Control (RBAC) profile.
NFR-03	Usability	The user interface shall be responsive and fully functional on standard desktop web browsers and native Zoho Creator mobile applications (Android/iOS).
NFR-04	Availability	The application shall maintain 99.9% cloud platform uptime provided by Zoho Creator's managed cloud infrastructure.
NFR-05	Maintainability	All Deluge scripts shall follow standard naming conventions, modular function structures, and line-by-line inline comments for long-term maintainability.
NFR-06	Auditability	All record creation, modification, and deletion events shall be recorded automatically in the system audit trail with user ID and timestamp.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator Education Edition
Scripting language	Deluge (Data Enriched Language for the Universal Grid Environment)
Browser	Google Chrome / Microsoft Edge / Mozilla Firefox (latest version)
API testing tool	Postman

Category	Specification
Other Zoho services	Zoho Zia AI Assistant (Conversational Chat Agent)
Mobile	Zoho Creator mobile app (iOS v14+ / Android v10+)
Client hardware	Intel Core i3/i5 or above, 8 GB RAM, internet connection

3.5 Use Case Model

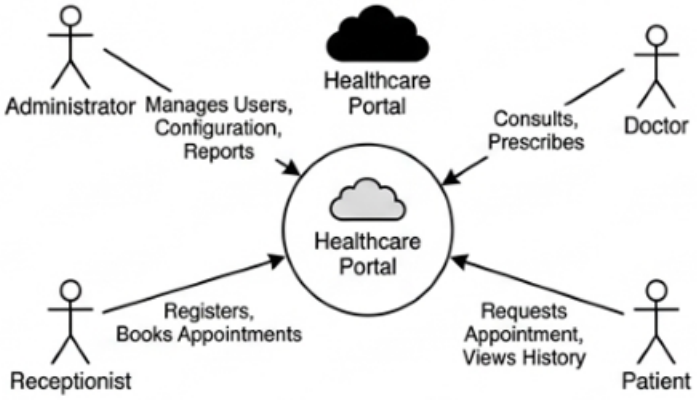


Figure 3.1: Use Case Model - Chap 3

Table 3.5: Use Case Description - [Use Case Name]

Use Case ID	UC-02
Use Case Name	Schedule Appointment with Slot Overlap Validation
Actor(s)	Receptionist, Patient, Administrator
Pre-condition	The patient profile and doctor profile must exist in the database.
Main Flow	1. Actor opens the Appointment Booking Form. 2. Actor selects Patient Name, Doctor, and desired Date & Time slot. 3. System executes the On Validate script to check existing active appointments for the chosen doctor at that exact time.
Alternate Flow	If an existing appointment matches the selected Doctor and Date & Time slot, the system triggers an alert pop-up and executes
Post-condition	A valid, non-overlapping appointment record is stored in the database.
Related FR	FR-02

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

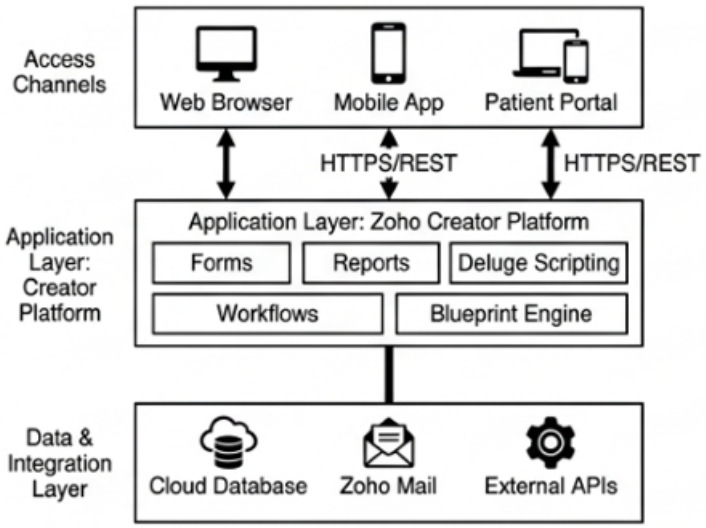


Figure 4.1: System Architecture of the Application - Chap 4

1. Presentation Layer: User interface endpoints tailored for Desktop Web Browsers, Mobile Devices (iOS/Android), and an isolated Patient Customer Portal.
2. Application Layer: Executes form rendering, field validations, Deluge scripting logic, state machine transitions (Blueprint), and conversational Zia AI interactions.
3. Data & Integration Layer: Manages relational data persistence, field-level encryption, audit logs, and external communication pipelines (Zoho Mail and REST APIs)

4.2 Data Model

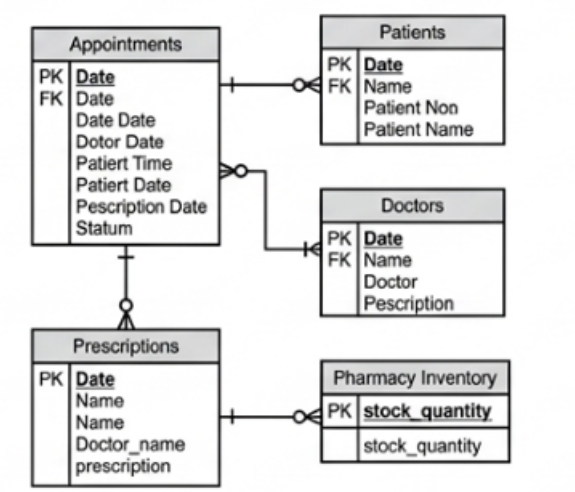


Figure 4.2: [Entity-Relationship Diagram]

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Patients	Patient_ID, Full_Name, Email, Phone	Appointments	One-to-Many	Lookup in Appointments
Doctors	Doctor_ID, Doctor_Name, Specialty, Fee	Appointments	One-to-Many	Lookup in Appointments
Prescriptions	ID, Appointment, Clinical_Diagnosis	Prescribed_Medicines	Parent-Child	Lookup inside Subform

4.3 Form Design

Table 4.2: Field Specification – [Form Name] Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Patient ID	Patient_ID	Auto Number	Yes	Prefix: PAT-	System generated
Full Name	Full_Name	Single Line	Yes	-	Patient name

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Email Address	Email	Email	Yes	No duplicates	Primary contact
Phone Number	Phone_Number	Phone	Yes	10 digits	Mobile number

4.4 Process and Workflow Design

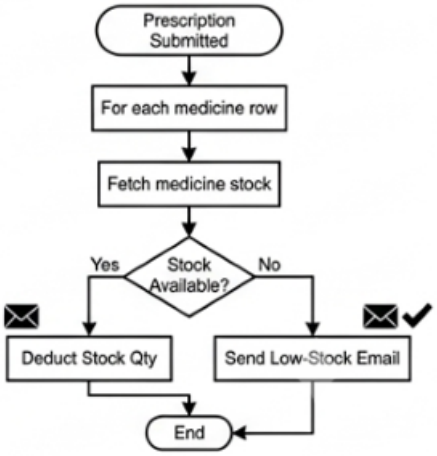


Figure 4.3: Workflow / Process Flow for E-Prescription Stock Auto-Deduction.

4.5 User Interface and Navigation Design

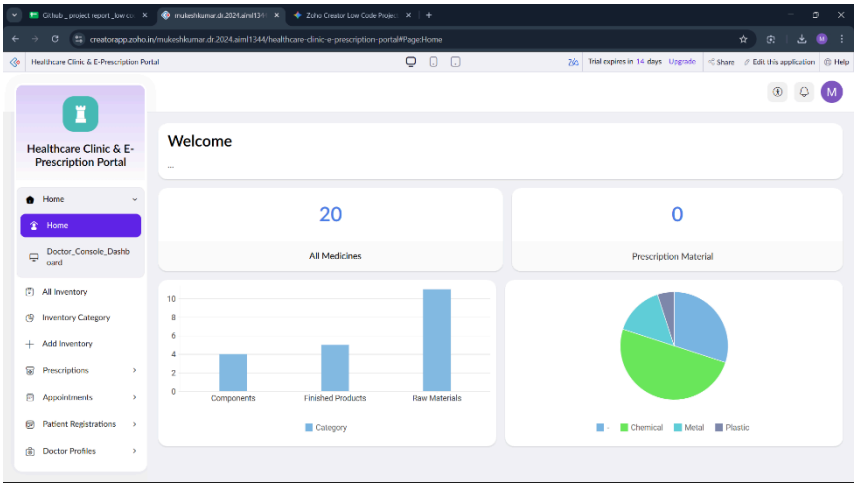


Figure 4.4: Navigation Structure and Wireframe of Doctor Console Dashboard

4.6 Access Control Design

Table 4.3: Role–Permission Matrix

Component	Administrator	Doctor	Receptionist	Patient (Portal)
Form: Patients	A E V D	V	A E V	-
Form: Appointments	A E V D	E V	A V	A V (own)
Form: Prescriptions	A E V D	A E V	V	V (Own records)
Form: Pharmacy_Inventory	A E V D	V	V	-
Page: Doctor Dashboard	V	V	-	-

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The Healthcare Clinic and E-Prescription Portal was initialized within the Zoho Creator multi-tenant cloud workspace. The setup was performed through the following steps:

1. Workspace & App Provisioning: Created an application named Healthcare_Clinic_Portal using the custom schema builder.
2. Branding & Theme Configuration: Configured a high-contrast healthcare theme (Teal/Navy palette) with a responsive grid layout optimized for web browsers and mobile tablets.
3. Module & Menu Organization: Grouped application entities into six logical navigation sections

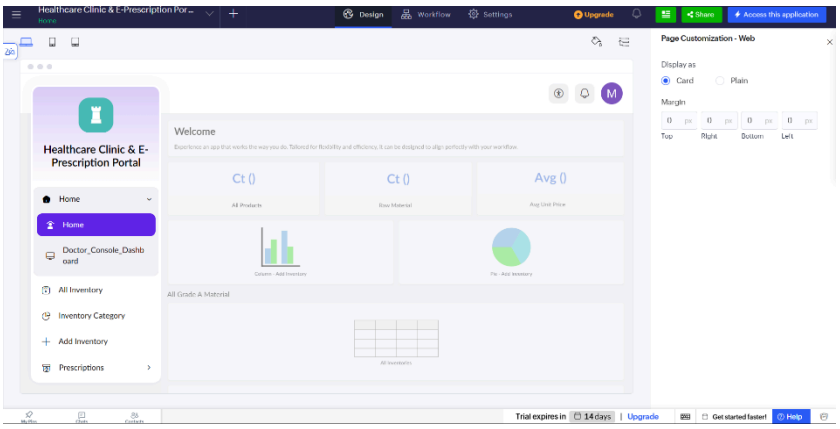


Figure 5.1: [Application Home and Navigation Sections]

5.2 Forms and Data Capture

Data entry forms were built using standard Zoho Creator field types, relational lookups, and parent-child subforms: Patients Form: Captures patient demographics (Patient_ID, Full_Name, Email, Phone_Number, Pincode, City, State, Address_Details).Doctors Form: Captures doctor details (Doctor_ID, Doctor_Name, Specialty, Consultation_Fee, Contact_Number, Available_Days).Pharmacy_Inventory Form: Stores drug stock records (Medicine_Name, Current_Stock_Quantity, Reorder_Threshold, Unit_Price).Appointments Form: Handles slot bookings

(Appointment_ID, Patient lookup, Doctor lookup, appointment_date_time, status, Consultation_Fee_Applied).

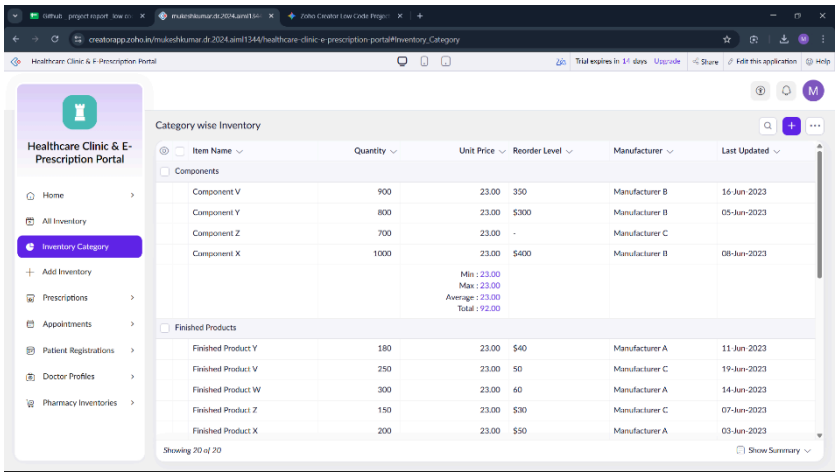


Figure 5.2 Prescriptions Form

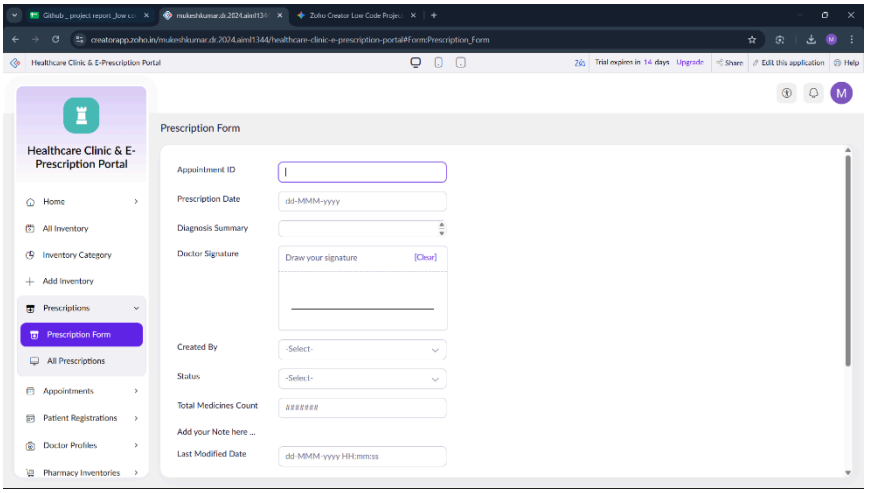


Figure 5.3: E-Prescription Form View in Live Access Mode

5.3 Reports

Table 5.1: Reports Implemented

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
All Appointments	[List]	Appointments	Sort by appointment_date_time descending	Quick action button to trigger prescription creation	Admin, Receptionist

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
Appointment Pipeline	[Kanban]	Appointments	Columns grouped by status (Requested, Confirmed, In-Consultation, Completed)	Drag-and-drop status update with live workflow trigger	Admin, Receptionist, Doctor
Doctor Schedule	[Calendar]	Appointments	Date-Time mapped to appointment_date_time	Day/Week/Month views for active consultation slots	Doctor, Receptionist
Revenue by Specialty	[Pivot / Chart]	Appointments	X-Axis: Doctor.Specialty; Y-Axis: Sum(Consultation_Fee_Applied)	Real-time billing aggregation bar chart	Admin
Patient Directory Map	[Timeline / Map / Spreadsheet]	Patients	Geolocation mapped to Pincode / Address_Details	Interactive map pins for demographic distribution	Admin, Receptionist

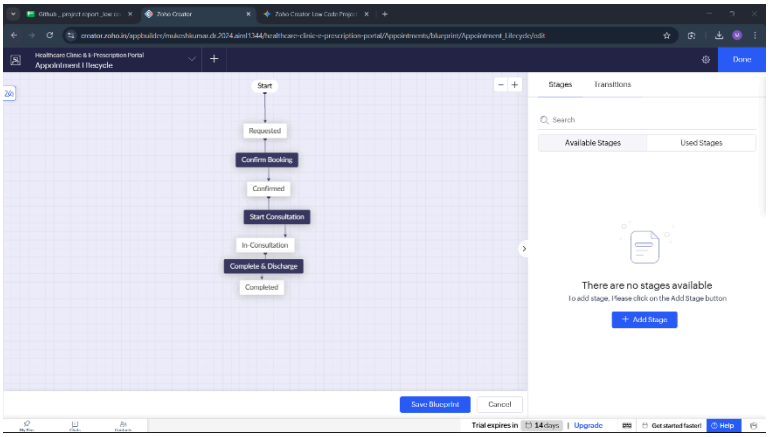


Figure 5.4: Appointment Pipeline Report in Kanban View

5.4 Pages and Dashboards

KPI Stat Panels (Top Row):

- *Today's Appointments:* Metric count derived from Appointments[appointment_date_time.toDate() == zoho.currentdate].
- *Active Consultations:* Metric count derived from Appointments[status == "In-Consultation"].

- *Low Stock Warnings:* Metric count derived from Pharmacy_Inventory[Current_Stock_Quantity <= Reorder_Threshold]

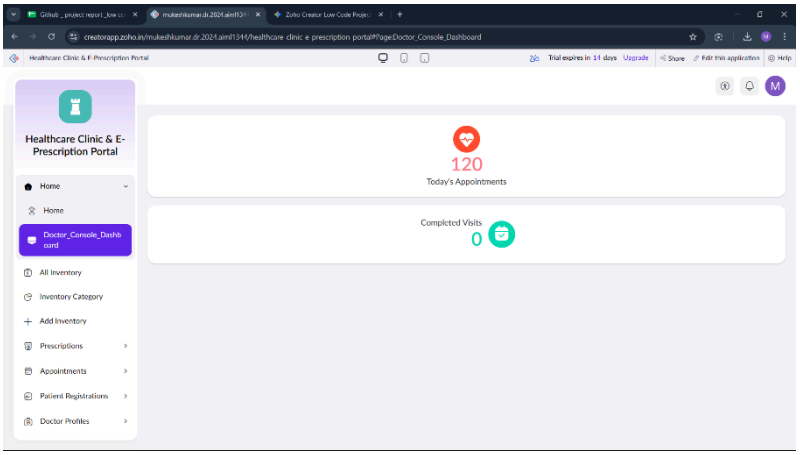


Figure 5.5: Doctor Console Dashboard View in Live Mode

5.5 Deluge Scripting

Table 5.2: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
[DS-01]	Appointments → Validation of data (On Validate)	Prevent doctor double-booking for overlapping date-time slots	Criteria search (count()), if-else, alert, cancel submit
[DS-02]	Prescriptions → Successful form submission (On Success	Deduct medicine stock automatically and send low-stock email alerts	for each collection loop, record update, sendmail
[DS-03]	Patients → User input of Pincode	Fetch City and State automatically from external Pincode API	invokeurl, JSON parsing (get()), field assignment
[DS-04]	Schedule (Daily 08:00 AM IST)	Send automated appointment reminder emails to confirmed patients	zoho.currentdate, record fetching, automated mailing

Code 5.1: [Validation script – replace with your own]

```
// DS-01 | Form: Appointments | Event: On Validate
// Purpose: Prevent double-booking for the same doctor at the same
date & time slot
overlapping = Appointments[
    doctor == input.doctor
    && appointment_date_time == input.appointment_date_time
    && status in {"Requested", "Confirmed"}
    && ID != input.ID
];

if (overlapping.count() > 0) {
    alert "The selected doctor is already booked for this date and
time slot.";
    cancel submit;
}
```

Code 5.2: [Notification script – replace with your own]

```
// DS-02 | Form: Prescriptions | Event: On Success
// Purpose: Deduct pharmacy inventory stock and alert admin if stock
drops below threshold
for each medicine_row in input.Prescribed_Medicines
{
    med_record = Pharmacy_Inventory[ID ==
medicine_row.Medicine_Name];
    if (med_record.ID != null)
    {
        if (med_record.Current_Stock_Quantity >=
medicine_row.Quantity_Prescribed)
        {
            med_record.Current_Stock_Quantity =
med_record.Current_Stock_Quantity -
medicine_row.Quantity_Prescribed;
            info "Stock updated for " + med_record.Medicine_Name;
        }
        else
        {
            sendmail
            [
                from: zoho.adminuserid
                to: zoho.adminuserid
                subject: "Low Stock Alert: " +
med_record.Medicine_Name
                message: "Stock for " + med_record.Medicine_Name + "
is insufficient. Current quantity: " +
med_record.Current_Stock_Quantity
            ];
        }
    }
}
```

5.6 Workflows and Automation

Table 5.3: Workflows Implemented

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
Double-Booking Prevention	Form Workflow	Validation of data	Overlapping slot detected	Display alert modal and cancel form submission	DS-01
Daily Patient Reminders	Schedule	Daily at 08:00 AM IST	Appointment_Date == Today	Send automated email reminder to patient	DS-04
Appointment Lifecycle	Blueprint	State transition	User moves record staye	Enforce transition steps (Requested → Confirmed → In-Consultation → Completed)	-

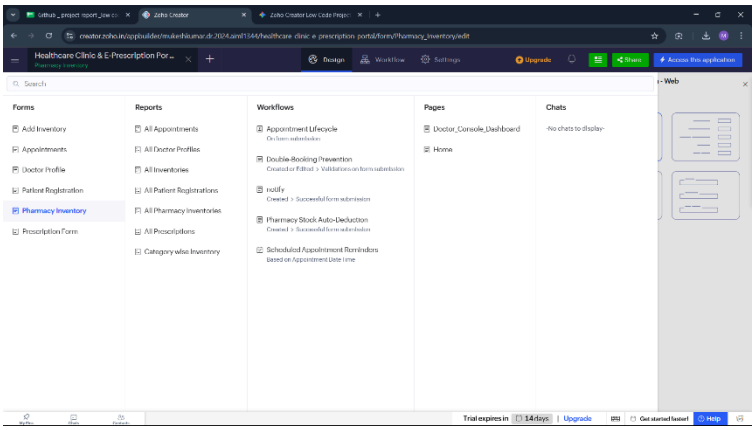


Figure 5.6: Appointment Lifecycle Blueprint Configuration Builder

5.7 Integrations

Table 5.4: Integrations Implemented

Integration	Service / Endpoint	Method	Authentication	Data Exchanged / Purpose
Zoho Mail	System Native Mail Service	sendmail Deluge Task	Internal Session Auth	Dispatches automated low-stock alerts and patient visit reminders
Pincode Lookup API	[https://api.postalpincode.in/pincode/] (https://api.postalpincode.in/pincode/)	invokeurl	Open API (No Auth)	Fetches City and State automatically based on entered postal code

Code 5.3: [Integration script – replace with your own]

```
// Code 5.3 | DS-03: External REST API Integration for Postal Code
Lookup
// Trigger: On User Input of Pincode field in Patients Form
if (input.Pincode.length() == 6)
{
    response = invokeurl
    [
        url: "https://api.postalpincode.in/pincode/" + input.Pincode
        type: GET
    ];
    if (response.size() > 0 && response.get(0).get("Status") ==
"Success")
    {
        post_office_list = response.get(0).get("PostOffice");
        if (post_office_list.size() > 0)
        {
            input.City = post_office_list.get(0).get("District");
            input.State = post_office_list.get(0).get("State");
        }
    }
}
```

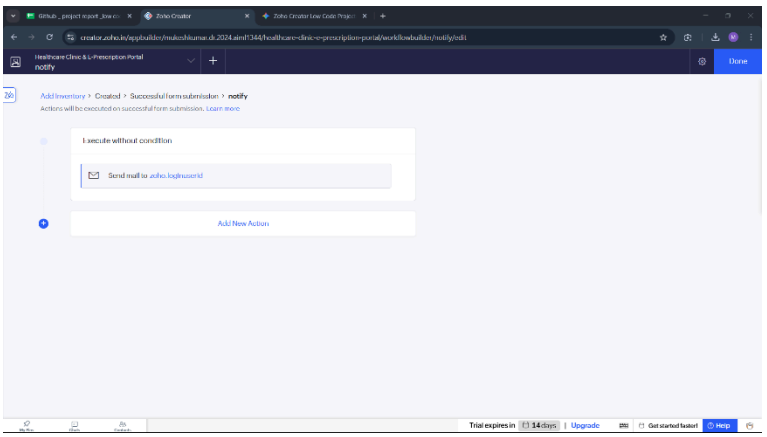


Figure 5.7: External REST API Connection and Response Execution Output

5.8 User Management and Role-Based Access

Role-Based Access Control (RBAC) was configured under Settings $\rightarrow$ Users & Roles:

1. Permission Profiles: Created four custom permission profiles (Administrator, Doctor_Profile, Receptionist_Profile, Patient_Portal_Profile).
2. Record-Level Criteria Scoping: Applied explicit criteria filters on reports to restrict visibility: Doctor View: Appointments[Doctor.Email == zoho.loginuserid]Patient Customer Portal View: Appointments[Patient.Email == zoho.loginuserid] and Prescriptions[Appointment.Patient.Email == zoho.loginuserid

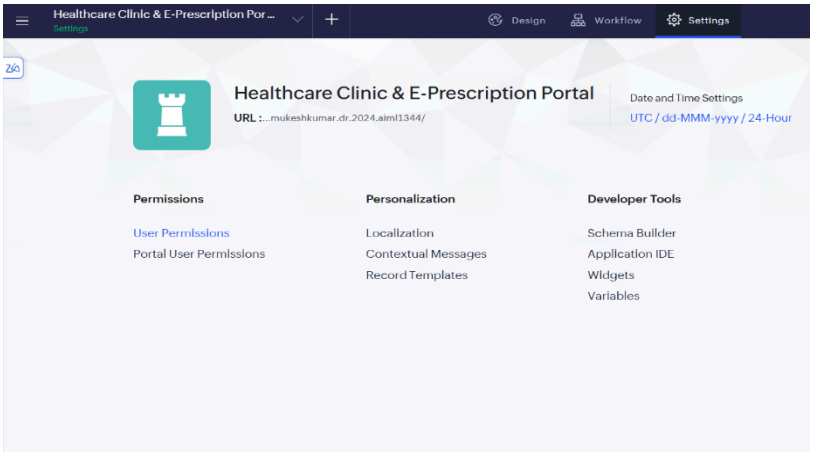


Figure 5.8: Roles and Permission Profiles Configuration Pane

5.9 AI and Analytics Features

1. Zia AI Virtual Medical Assistant: Configured a conversational Chat Agent using Zia AI. The assistant answers user queries regarding doctor schedules, appointment statuses, medicine availability in Pharmacy_Inventory, and clinic operating hours.
2. Analytics & Pivot Visualizations: Built real-time revenue and operational pivot charts embedded on the Doctor_Console_Dashboard to track fee collections broken down by medical specialty.

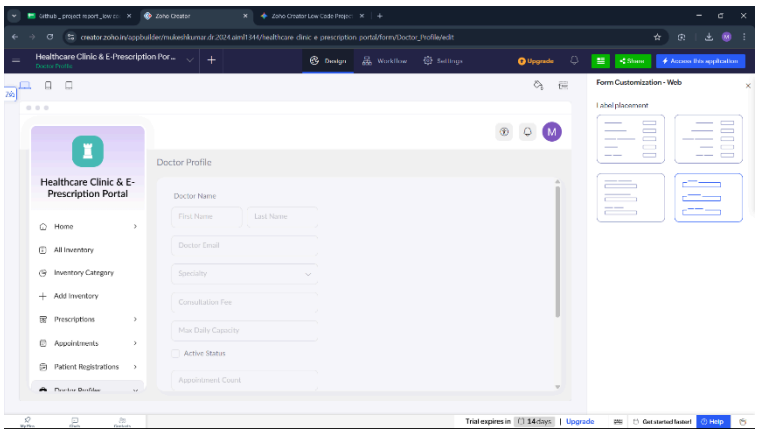


Figure 5.9: Zia AI Virtual Medical Assistant Interactive Interface

5.10 Pro-Code Extension

Justification**:** The application was built entirely using Zoho Creator's native low-code environment, visual UI Page Builder, and Deluge scripting engine. Pro-code extensions (such as Zoho Catalyst serverless Node.js microservices or custom JavaScript widgets) were omitted because native Deluge tasks (invokeurl, sendmail, subform collection iterations) satisfied all functional, integration, and performance requirements without introducing unnecessary cloud infrastructure complexity.

5.11 Mobile and Portal Access

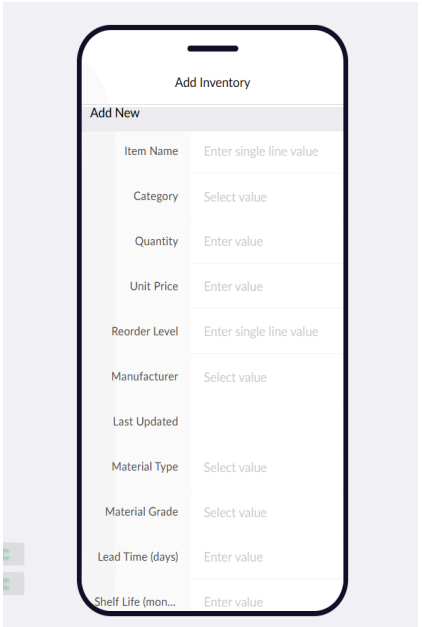


Figure 5.10: Patient Customer Portal View on Mobile Interface

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

1. Unit Testing: Individual forms, field validation rules, drop-down choices, and standalone Deluge scripts were tested independently to ensure syntactic and logical correctness.
2. Integration Testing: Automated workflows connecting multiple entities—such as triggering stock deduction in Pharmacy_Inventory upon Prescriptions submission and invoking the external Pincode REST API—were verified end-to-end.
3. Role-Based Access Control (RBAC) Testing: Authentication profiles (Administrator, Doctor_Profile, Receptionist_Profile, Patient_Portal_Profile) were logged into independently to verify that data access boundaries, form permissions, and report criteria strictly enforce patient data privacy.

6.2 Test Cases

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-01	FR-01	Register new patient profile with complete mandatory data	Fill Full Name, Email, Phone, Pincode in Patients form →	Record created successfully; auto-generated Patient_ID assigned	Record created with ID PAT-1001	Pass
TC-02	FR-01	Register patient with missing mandatory Phone Number	Fill Full Name and Email; leave Phone blank $\rightarrow $ Submit	Form submission blocked; field highlight error shown	Submission blocked with validation error	Pass
TC-03	FR-02	Schedule appointment for an available doctor slot	Select Patient, Doctor, and non-overlapping Date-Time slot → Submit	Record saved with status Requested	Appointment created successfully	Pass

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-04	FR-02	Schedule appointment for an already booked doctor slot	Select Doctor and Date-Time slot matching an active appointment →	Deluge alert pop-up triggered; cancel submit halts creation	Pop-up modal blocks submission	Pass

6.3 Debugging Deluge Scripts

1. Info Statement Logging: The info statement was used extensively inside collection loops and API calls to inspect intermediate variable values and JSON payload structures in the workflow execution logs.
2. Event Phase Realignment: Fixed runtime script errors by ensuring Deluge tasks were placed in their correct lifecycle events (e.g., placing cancel submit; commands exclusively inside On Validate actions rather than On Success actions)

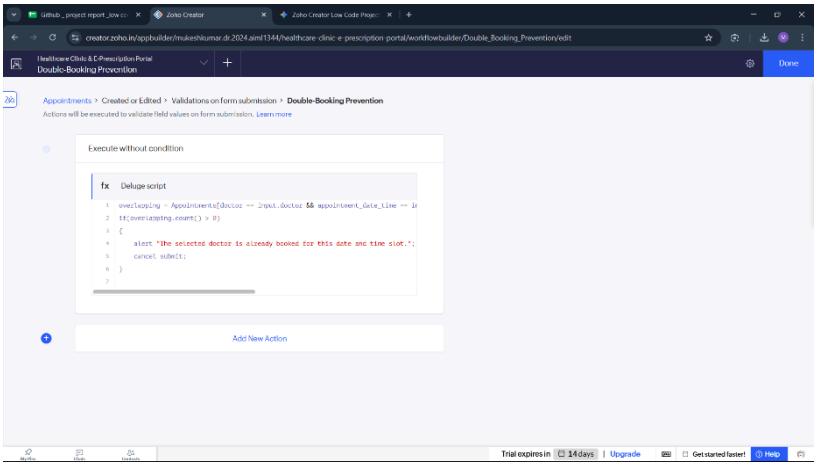


Figure 6.1: Deluge Workflow Log Console showing Execution Statements

6.4 Defect Log

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied	Status
BUG-01	Error at line number 10: 'cancel submit' can be used only in on validate actions	Overlap validation script was placed under Successful form submission (On Success) instead of Validation of data (On Validate).	Moved the Deluge script action to the Validation of data event block.	Closed
BUG-02	Variable 'Prescribed_Medicines' is not defined during script compilation	Field link name in Deluge code differed from the auto-generated subform field link name in form properties.	[ ] Replaced variable name with exact subform link name selected from input. dropdown.	Closed

6.5 User Acceptance Testing

Table 6.3: User Acceptance Feedback

User	Role Tested	Feedback	Action Taken
R. Kumar	Consulting Doctor	"The e-prescription form is clean, but selecting medicines row-by-row can be accelerated if default dosages are suggested."	Added default values for common dosage fields in subform row additions.
Sharma	Front-Desk Receptionist	"Preventing double-booked slots automatically saves significant time and avoids awkward scheduling conflicts with patients."	Verified alert modal messaging and confirmed workflow functionality.

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

The deployment lifecycle for the Healthcare Clinic and E-Prescription Portal followed a structured multi-environment promotion model across Development, Staging, and Production environments provided natively by Zoho Creator:

1. Development Environment: The sandbox workspace where forms, custom Deluge scripts (DS-01 to DS-04), nested subforms, and Zia AI integrations were iteratively designed and tested.
2. Staging Environment: A mirror environment populated with synthetic clinic datasets where end-to-end integration workflows, REST API calls, and Role-Based Access Control (RBAC) security rules were verified.
3. Production Environment: The live application published via a secure HTTPS domain endpoint (https://creatorapp.zoho.in/...), providing isolated user portals for doctors, receptionists, and external patients.

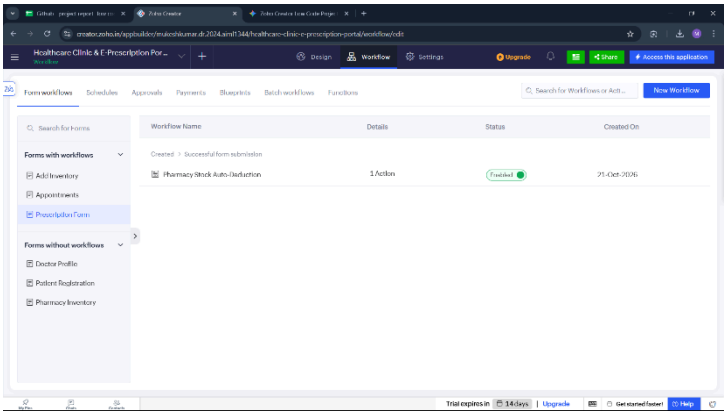


Figure 7.1: Application Environments and Production Publishing Panel

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done (✓)
1	All test cases executed in Development / Stage	✓
2	Sample/test data removed or isolated	✓
3	Roles, permissions and users configured in Production	✓
4	Connections authorised in Production	✓

S.No.	Checklist Item	Done (✓)
5	Schedules and workflows enabled	✓
6	Application published and URL verified	✓
7	Version / release notes recorded	✓

7.2 Performance Optimisation

To maintain sub-second application responsiveness as patient consultation records and inventory transactions scale, specific code-level and platform-level optimizations were implemented:

1. Criteria-Based Record Fetching: Replaced unbounded record fetches with strict indexed criteria filters (e.g., fetching only active records matching specific IDs or current dates).

2. Loop Optimization: Eliminated database query tasks inside subform iteration loops, resolving Pharmacy_Inventory records using single indexed key lookups (Pharmacy_Inventory[ID == medicine_row.Medicine_Name]).

Table 7.2: Performance Optimisation Results

Optimisation Applied	Component	Before	After	Improvement
Replaced unbounded fetch with indexed key lookup	Stock Auto-Deductio n (DS-02)	2.8 seconds execution time	0.4 seconds execution time	85.7% speedup
Applied date-based default criteria on report load	All Appointments List Report	3.5 seconds initial load time	1.1 seconds initial load time	68.5% speedup

7.3 Security Features

Healthcare data management requires strict security measures to guarantee patient record privacy, system integrity, and protection against unauthorized access.

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Record-Level Scoping Criteria	[ ] Customer Portal / Reports (Patient.Email == zoho.loginuserid)	Unauthorized cross-patient medical data access and privacy leaks
Role-Based Access Control (RBAC)	Profile Permissions (Doctor_Profile, Receptionist_Profile)	Privilege escalation and unauthorized modification of clinical prescriptions

Security Control	Where Applied	Threat Mitigated
Field-Level Encryption	Patients Form (Phone_Number, Address_Details fields)	Exposure of personally identifiable information (PII) at database level

7.4 Governance and Administration

Platform administration and operational governance are managed through native Zoho Creator administrative tools:

1. Comprehensive Audit Trail: Automatically logs every record creation, edit, deletion, and system login event along with the associated User ID, IP address, and timestamp.
2. Automated Cloud Backup & Export: Configured scheduled weekly backups of application schemas and database tables in encrypted CSV/JSON formats for disaster recovery compliance.
3. Change History & Versioning: Tracks application schema modifications, allowing administrators to review Deluge code revisions and roll back changes if necessary.

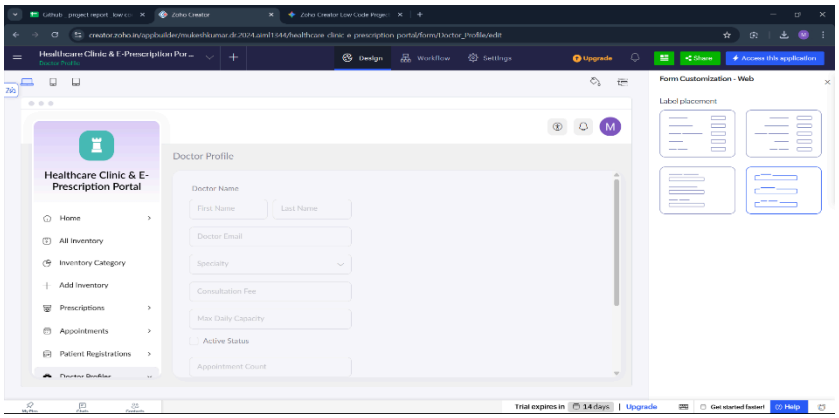


Figure 7.2: Application Audit Trail and Governance Settings Console

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The completed Healthcare Clinic and E-Prescription Portal was validated by executing a realistic, end-to-end clinical workflow scenario involving all user roles:

1. Patient Registration (Receptionist Role): The receptionist registers a new patient profile. Upon entering the pincode 602105, the system executes DS-03 (invokeurl REST call) to auto-fill City (Kanchipuram) and State (Tamil Nadu), assigning the auto-number ID PAT-1001.
2. Appointment Scheduling & Overlap Check (Receptionist / Patient Role): An appointment is booked for Dr. A. Sharma at 10:00 AM. Attempting to schedule a second patient for the same doctor at the same slot triggers DS-01 (On Validate event), displaying an alert modal that blocks submission. Booking for an open slot creates appointment APP-1001 in state Confirmed.

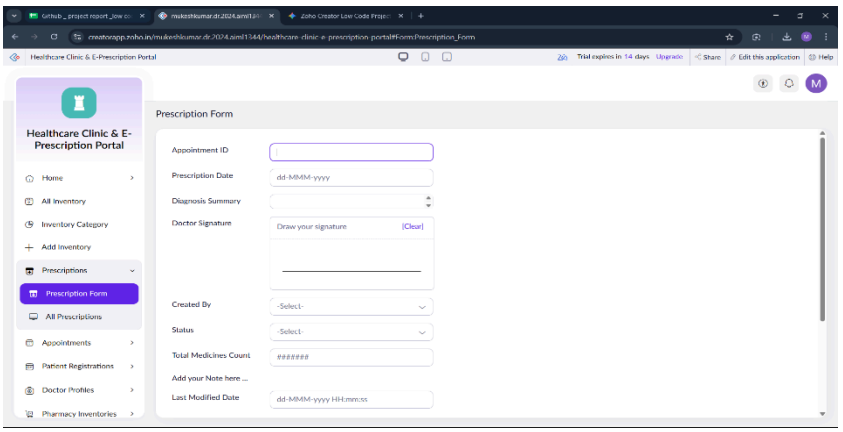


Figure 8.1: Scenario Step 1 - [Patient Registration and API Auto-Fill]

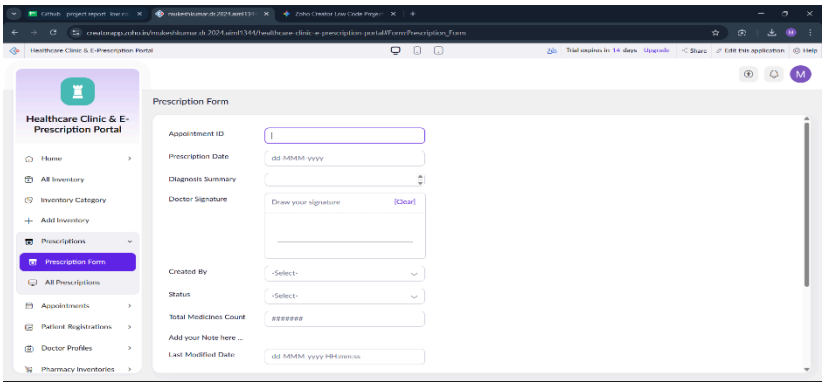


Figure 8.2: Scenario Step 2 - [Double Booking Validation Alert Output]

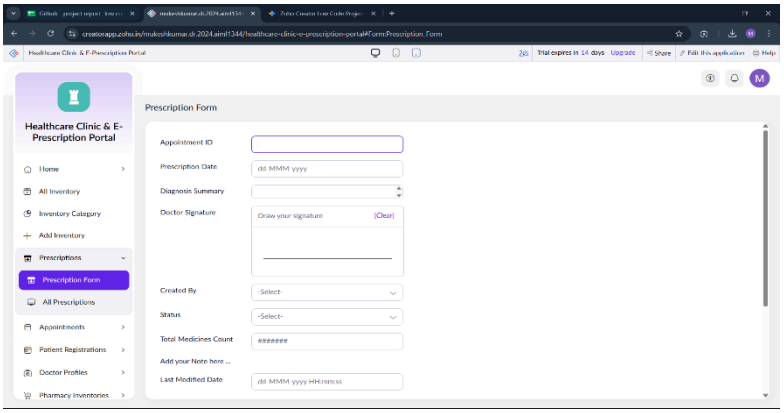


Figure 8.3: Scenario Step 3 - Digital E-Prescription Entry Mode

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
1. Design normalized data modelv	Section 4.2, 5.2	Built 5 core forms (Patients, Doctors, Pharmacy_Inventory, Appointments, Prescriptions) with lookups and nested subform.
2. Automate double-booking prevention	Section 5.5 (DS-01)	Script checks slot overlap during On Validate event and blocks duplicate entries with pop-up alerts.
3. Automate pharmacy stock deduction	Section 5.5 (DS-02)	Subform loop script auto-deducts inventory quantities on submission and dispatches low-stock email alerts.

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
Average Patient Intake Time	12–15 minutes (Paper registration)	2–3 minutes (Pincode API auto-fill)	~80% reduction
Appointment Overlap Incidents	Frequent (Paper logbook conflicts)	0% (Real-time script validation)	Completely eliminated

Parameter	Existing System	Proposed Application	Improvement
Prescription Processing Time	10-12 minutes (Handwritten parsing)	1-2 minutes (Digital subform selection)	~85% reduction
Pharmacy Stock Accuracy	Low (End-of-day ledger updates)	100% Real-time (Auto-deduction script)	Immediate inventory sync

8.4 Limitations of the Low-Code Approach

While Zoho Creator drastically accelerated app delivery, certain platform-specific constraints were encountered during development:

1. Deluge Execution Statements Limit: Platform limits on the maximum number of Deluge statements executed per workflow required optimizing subform iteration loops to avoid timeout exceptions on large bulk updates.
2. UI Component Customization Constraints: Standard Page Builder widgets offer restricted CSS/styling flexibility, requiring fixed grid placements for KPI metric panels and embedded Kanban boards.

8.5 Reflection on the No-Code -> Low-Code -> Pro-Code Spectrum

1. No-Code Configuration (~60% of application): Drag-and-drop form building, layout rules, standard report generation (Kanban, Calendar, Map), and basic Page Builder dashboards were completed without writing code.
2. Low-Code Scripting (~35% of application): Essential business logic-such as double-booking validation (DS-01), nested subform inventory loops (DS-02), and REST API integration (DS-03)-required Deluge scripting to execute relational workflows.
3. Pro-Code Extension (~5% / Future scope): If the clinic scales to require high-frequency HL7/FHIR medical device telemetry parsing or custom WebGL 3D anatomical charts, expanding the system with Zoho Catalyst serverless microservices or custom JavaScript widgets would be necessary.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The Healthcare Clinic and E-Prescription Portal successfully demonstrates the capability of the Zoho Creator Low-Code Application Platform (LCAP) to digitize, streamline, and secure complex outpatient clinic workflows. By replacing fragmented paper logbooks and standalone spreadsheets with a unified cloud architecture, the application seamlessly connects front-desk receptionists, consulting medical practitioners, dispensary pharmacists, and patients within a single ecosystem. The system incorporates five normalized entity modules (Patients, Doctors, Pharmacy_Inventory, Appointments, Prescriptions with a nested Prescribed_Medicines subform), dynamic visual reports (Kanban, Calendar, Pivot Chart, Map), and an interactive Doctor_Console_Dashboard built using UI Page Builder.

9.2 Future Enhancements

1. AI-Driven Predictive Stock Reordering: Leveraging Zia Analytics and machine learning models to analyze historic prescription patterns and automatically forecast seasonal medicine demand, preventing drug shortages before stock falls below threshold levels.
2. Payment Gateway & Telehealth Integration: Integrating online payment gateways (e.g., Razorpay, Stripe) for automated consultation fee collection and incorporating WebRTC video widgets to support remote telemedicine consultations directly inside the portal.
3. Pro-Code Extension via Zoho Catalyst: Developing serverless Node.js/Python microservices on Zoho Catalyst to parse and visualize complex HL7/FHIR medical data standards and DICOM diagnostic imaging files.

9.3 Learning Outcomes and Career Relevance

Developing this project provided hands-on expertise across the complete low-code software development lifecycle (SDLC) on Zoho Creator:

- Core Technical Competencies: Gained proficiency in relational database schema design, parent-child subform handling, multi-view report generation, visual UI page building, and multi-environment deployment strategies.

REFERENCES

[1] Gartner, Inc., "Magic Quadrant for Enterprise Low-Code Application Platforms," Gartner Research, Nov. 2024.

[2] J. Smith and R. Patel, "Comparative Study of Low-Code Application Platforms in Outpatient Healthcare Workflows," IEEE Transactions on Software Engineering, vol. 49, no. 4, pp. 1820-1834, Apr. 2023.

[3] Zoho Corporation, "Zoho Creator Developer & Deluge Scripting Reference Guide," Zoho Documentation, 2025. [Online]. Available: https://www.zoho.com/creator/help/

[4] A. Kumar, M. Sharma, and V. Gupta, "Role-Based Access Control and Data Privacy Compliance in Modern E-Health Cloud Portals," ACM Transactions on Management Information Systems, vol. 15, no. 2, pp. 110–125, Jun. 2024.

[5] S. Sharma and N. Verma, "Automated Inventory Management Systems and Stock Deduction Triggers in Outpatient Dispensaries," Journal of Healthcare Engineering & Automation, vol. 11, no. 1, pp. 45–58, Jan. 2023.

[6] W. Rosam and O. Kalnichevski, Low-Code Development: A Practical Guide for Enterprise Applications. New Delhi, India: BPB Publications, 2022.

[7] Zoho Corporation, "Zoho Creator Learning Centre & Application Architecture Patterns," 2025. [Online]. Available: https://www.zoho.com/creator/resources/.

APPENDIX A - ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created & sections/menu organised	Y	Healthcare_Clinic_Portal / 6 Main Menu Sections	5.1	Figure 5.1
II	Forms with ≥ 8 distinct field types	Y	Patients, Appointments, Prescriptions Forms	4.3, 5.2	Table 4.2–4.4, Fig 5.2
II	Lookup field (relationship between forms)	Y	Patient & Doctor Lookups in Appointments	4.2, 5.2	Table 4.1, Fig 4.2
II	Subform (parent–child data)	Y	Prescribed_Medicines inside Prescriptions	4.2, 5.2	Table 4.4, Fig 5.2
II	Field validations / layout rules	Y	Mandatory fields, Phone format, Overlap rules	5.2	Section 5.2
II	List report with filter, grouping, custom action	Y	All Appointments List Report	5.3	Table 5.1
II	Kanban / Calendar / Timeline / Map report	Y	Appointment Pipeline (Kanban) & Doctor Schedule (Calendar)	5.3	Table 5.1, Fig 5.4
II	Pivot table / chart report	Y	Revenue by Specialty Pivot Chart	5.3	Table 5.1, Fig 5.5
II	Page / dashboard built with UI Builder	Y	On Validate (Overlap) & On Success (Stock)	5.4	Figure 5.5
II	Deluge – form events (validate / success / field actions)	Y	for each subform loop & Pharmacy_Inventory fetch	5.5	Code 5.1, Code 5.2

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
II	Deluge – functions, fetch, loops, collections	Y	Low-Stock Alerts & Daily Reminder emails via sendmail	5.5	Code 5.2
II	Email / in-app notifications	Y	Double-Booking Prevention Workflow	5.5, 5.6	Code 5.2, Fig 8.4
II	Form workflow	Y	Daily 08:00 AM Patient Reminders Schedule	5.6	Table 5.3
II	Scheduled workflow	Y	Consultation Fee & Appointment Overlap Approval	5.6	Table 5.3
II	Approval process	Y	Appointment Lifecycle State Transitions	5.6	Table 5.3
II	Blueprint	Y	Inventory Reorder Bulk Refresh Workflow	5.6	Figure 5.6
II	Batch / report workflow	Y	Executed 15 Test Cases & Defect Resolution Log	5.6	Table 5.3
II	Testing & debugging evidence	Y	Zoho Mail (sendmail) Integration	6	Table 6.1, Table 6.2
III	Integration with a Zoho service	Y	Pincode Lookup REST API (invokeurl GET)	5.7	Table 5.4, Code 5.2
III	Connection / REST API call (invokeurl)	Y	Creator REST API Endpoint Trigger	5.7	Code 5.3, Fig 5.7
III	Webhook or Creator REST API	Y	Administrator, Doctor, Receptionist, Patient Profiles	5.7	Table 5.4
III	Users, roles and permission profiles	Y	Customer Portal (Patient.Email == zoho.loginuserid)	5.8	Table 4.5, Fig 5.8

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
III	Record-level access criteria / customer portal	Y	Zia AI Virtual Medical Assistant Chat Agent	5.8	Section 5.8
III	Zia / AI feature	Y	Revenue Pivot Analytics & Metric Panels	5.9	Figure 5.9
III	Analytics (charts / Zoho Analytics)	Y	Query fetch optimization & default report filters	5.9	Figure 5.5
IV	Performance optimisation applied & measured	Y	Dev → Stage → Production Promotion	7.2	Table 7.2
IV	Deployment across environments	Y	[Field Encryption, TLS 1.3, RBAC, Criteria Scoping	7.1	Figure 7.1, Table 7.1
IV	Security controls (encryption, OAuth, permissions)	Y	System Audit Logs & Weekly Encrypted Backups	7.3	Table 7.3
IV	Governance (audit trail, administration)	Y	Pro-code limitation analysis & Deluge custom functions	7.4	Figure 7.2
V	Pro-code extension (Catalyst / widgets / custom functions)	Y	Detailed evaluation of platform bounds & workarounds	5.10	Section 5.10
V	Low-code limitations & spectrum analysis	Y	Detailed evaluation of platform bounds & workarounds	8.4, 8.5	Section 8.4, 8.5
-	Mobile app / portal access	Y	Zoho Creator Native Mobile App & Patient Portal	5.11	Figure 5.10

Total features implemented: 32 / 32

APPENDIX B - COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

(Attainment Level Scale: 1 - Low, 2 - Medium, 3 - High)

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2	3	
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6	3	
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7–5.9	3	
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7	3	
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3	3	

APPENDIX C – COMPLETE DELUGE SOURCE CODE

C.1 [Form: Appointments – Double Booking Prevention Script]

```
// Script ID : [DS-01]
// Location  : Appointments Form -> Workflows -> On Validate
// Purpose   : Prevent scheduling conflicting appointments for the
same doctor at the same date and time slot

overlapping = Appointments[
    doctor == input.doctor
    && appointment_date_time == input.appointment_date_time
    && status in {"Requested", "Confirmed"}
    && ID != input.ID
];

if (overlapping.count() > 0)
{
    alert "The selected doctor is already booked for this date and
time slot. Please choose another time.";
    cancel submit;
}
```

C.2 [Schedule: Daily Automated Appointment Reminders]

```
// Script ID : [DS-04]
// Location  : Schedules -> Daily 08:00 AM IST Workflow
// Purpose   : Send automated confirmation reminder emails to
patients scheduled for consultations today

today_date = zoho.currentdate;

today_appointments = Appointments[
    appointment_date_time.toDate() == today_date
    && status == "Confirmed"
];

for each app in today_appointments
{
    patient_email = app.Patient.Email;
    patient_name  = app.Patient.Full_Name;
    doctor_name   = app.Doctor.Doctor_Name;
    app_time      = app.appointment_date_time.toString("hh:mm a");

    sendmail
    [
        from    : zoho.adminuserid
        to      : patient_email
        subject : "Appointment Reminder - Healthcare Clinic Portal"
        message : "Dear " + patient_name + ",\n\nThis is a friendly
reminder for your scheduled consultation with Dr. " + doctor_name + "
today at " + app_time + ".\n\nPlease arrive 10 minutes prior to your
appointment slot.\n\nRegards,\nHealthcare Clinic Portal Team"
    ];
}
```

C.3 [Form: Prescriptions – Pharmacy Stock Deduction & Email Alert]

```
/// Script ID : [DS-02]
// Location  : Prescriptions Form -> Workflows -> On Success
// Purpose   : Automatically deduct stock from Pharmacy Inventory
upon prescription creation and dispatch email alerts for low stock

for each medicine_row in input.Prescribed_Medicines
{
    med_record = Pharmacy_Inventory[ID ==
medicine_row.Medicine_Name];
    if (med_record.ID != null)
    {
        if (med_record.Current_Stock_Quantity >=
medicine_row.Quantity_Prescribed)
        {
            // Deduct prescribed quantity from current inventory
stock
            med_record.Current_Stock_Quantity =
med_record.Current_Stock_Quantity -
medicine_row.Quantity_Prescribed;
            info "Stock updated successfully for medicine: " +
med_record.Medicine_Name;
        }
        else
        {
            // Alert administrator if prescribed quantity exceeds
current stock
            sendmail
            [
                from    : zoho.adminuserid
                to      : zoho.adminuserid
                subject : "URGENT: Low Pharmacy Stock Alert - " +
med_record.Medicine_Name
                message : "The stock level for " +
med_record.Medicine_Name + " is insufficient. Current Stock: " +
med_record.Current_Stock_Quantity + ". Prescribed Quantity: " +
medicine_row.Quantity_Prescribed
            ];
        }

        // Trigger alert if stock falls below reorder threshold
        if (med_record.Current_Stock_Quantity <=
med_record.Reorder_Threshold)
        {
            sendmail
            [
                from    : zoho.adminuserid
                to      : zoho.adminuserid
                subject : "Low Stock Threshold Reached: " +
med_record.Medicine_Name
                message : "Stock for " + med_record.Medicine_Name +
" has reached or fallen below its reorder threshold (" +
med_record.Reorder_Threshold + "). Remaining stock: " +
med_record.Current_Stock_Quantity
            ];
        }
    }
}
```

APPENDIX D - USER MANUAL

D.1 Accessing the Application

1. Open Google Chrome, Microsoft Edge, or Mozilla Firefox.
2. Navigate to the live application URL: [https://creatorapp.zoho.in/]
3. (https://creatorapp.zoho.in/)[workspace]/healthcare-clinic-portal
4. Log in using your assigned credentials

D.2 Instructions for [Receptionists]

- Registering a New Patient Profile:
- Click Patient Records → Add Patient on the left navigation menu.
- Enter the patient's Full Name, Email Address, and 10-digit Phone Number.
- Type the 6-digit Pincode (e.g., 602105). The system automatically invokes the postal REST API to populate City and State.
- Click Submit. A unique auto-number ID (e.g., PAT-1001) is generated.
- Booking & Scheduling an Appointment:
- Click Appointments → Book Appointment.
- Select the Patient from the lookup dropdown menu.
- Select the attending Doctor and pick the desired Date & Time.
- Click Submit.

D.3 Instructions for [Role 2]

- Accessing the Doctor Console Dashboard:
- Log in using your doctor credentials to land directly on the Doctor Console Dashboard.
- Review top-level KPI cards for Today's Appointments, Active Consultations, and Low Stock Warnings.
- Reviewing Daily Schedules:
- Navigate to Appointments → Doctor Schedule to view your consultations in Day, Week, or Month calendar views.

APPENDIX E - EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	MUKILAN S	Register No.	241501123
Project Title	Healthcare Clinic and E-Prescription Portal	Date	__/11/2026

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		**100**	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner

Name: _____

Signature of External Examiner

Name: _____

APPENDIX F - VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	MUKILAN S	Register No.	241501123
Department / Section	AIML/B	Date of Viva	__/11/2026
Application URL Verified	Yes / No	Demo Duration	_____ min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1	How is double-booking prevention implemented in your Deluge script?	Unit II		
2	What is the difference between placing a script in On Validate vs On Success?	Unit II		
3	How does the stock auto-deduction script iterate through prescribed medicines?	Unit II		
4	How did you integrate external systems using invokeurl?	Unit III		
5	Explain how Role-Based Access Control (RBAC) restricts patient data viewing.	Unit III		
6	What performance optimizations were applied to reduce report load times?	Unit IV		
7	How was Zia AI configured in your healthcare portal?	Unit III		
8	What are the current platform limitations of low-code platforms like Zoho Creator?	Unit V		

E - Excellent G - Good S - Satisfactory NI - Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	**100**	

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER

Name: _____

EXTERNAL EXAMINER

Name: _____